

Operations on Data Structures

Q. Discuss the various fundamental operations that can be performed on data structures, explaining each with a suitable example.

****Definition to Remember****

****Opera**
operatio
the effi

The Six Fundamental Operations

| Operation | Definition | Example |

| :--- | :-

Traversal Algorithm Example (Array)

text

Algorith

Trade-offs Between Data Structures

| DS | Fast Operations | Slow Operations |

| :--- | :-

****Must-Write Points (for 10 marks)****

1. Six fu

*****Quick Recall*****

*`Travers
(combin*

Algorithmic Implementations & Executable C Codes

1. Array Operations: Traversal, Insertion & Deletion (C Code)

```
#include <stdio.h>
```

```
void traverse(int arr[], int n) {
```

printf("A

```
int insertElement(int arr[], int n, int capacity, int element, int pos) {
```

if (n >=

```
int deleteElement(int arr[], int n, int pos) {
```

if (pos <

```
int main() {
```

int arr[1

```
printf("--- Initial Array ---\n");
```

traverse

```
printf("\n--- Inserting 25 at Index 2 ---\n");
```

n = inse

```
printf("\n--- Deleting Element at Index 3 (30) ---\n");
```

```
return 0;
```

```
---
```

2. Searching Algorithms: Linear Search vs. Binary Search (C Code)

```
#include <stdio.h>
```

```
// Linear Search - O(n)
```

```
// Binary Search - O(log n) [Requires Sorted Array]
```

```
int main() {
```

```
int linRes = linearSearch(arr, n, target);
```

```
int binRes = binarySearch(arr, n, target);
```

```
return 0;
```

```
---
```

3. Sorting & Merging Operations (C Code)

```
#include <stdio.h>
```

```
// Bubble Sort Operation - O(n^2)
```

```
// Merging Two Sorted Arrays - O(m + n)
```

```
void me
```

```
int main() {
```

```
int A[] =
```

```
mergeArrays(A, m, B, n, C);
```

```
printf("N
```

```
return 0;
```

```
}
```